LAPORAN PRATIKUM JARINGAN KOMPUTER
TCP/IP MODEL



OLEH

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2026

A. TUJUAN PEMBELAJARAN

1. Mahasiswa mampu melakukan identifikasi transmisi data menggunakan model TCP/IP.
2. Mahasiswa mampu melakukan identifikasi layer dari model TCP/IP.
3. Mahasiswa mampu menggunakan aplikasi untuk mengidentifikasi data dari model TCP/IP.

B. ALAT DAN BAHAN

1. Modul praktikum TCP/IP Model.
2. PC yang dilengkapi aplikasi Wireshark.
3. Jaringan aktif.
4. Lembar laporan praktikum.
5. Koneksi LAN dan Internet.
6. Terminal/Command Prompt.
7. Web browser.

C. DASAR TEORI

1. OSI Layer

OSI (Open System Interconnection) merupakan model referensi komunikasi yang digunakan untuk membahas spesifikasi protokol. Model OSI terdiri atas tujuh layer: Application, Presentation, Session, Transport, Network, Data Link, dan Physical.

Layer	Nama	Unit Enkapsulasi	Deskripsi/Fungsi
7	Application	Data	Mendefinisikan pelayanan komunikasi pada suatu aplikasi.
6	Presentation	Data	Mendefinisikan format data.
5	Session	Data	Mendefinisikan memulai, pengontrolan, dan mengakhiri komunikasi.
4	Transport	Segment	Mendefinisikan jenis pengiriman.
3	Network	Paket	Mendefinisikan pengiriman dari titik ke titik.
2	Data Link	Frame	Mendefinisikan cara pengaturan pengiriman data.
1	Physical	bit	Mendefinisikan bentuk interface dari sebuah media transmisi

2. TCP/IP Model

TCP/IP (Transmission Control Protocol/Internet Protocol) adalah gabungan protokol yang mengatur komunikasi data dalam proses pertukaran data dari satu komputer ke komputer lain

dalam jaringan Internet. TCP/IP merupakan protocol suite. Menurut modul, arsitektur TCP/IP menggunakan model referensi DARPA/DOD dan terdiri atas empat lapisan.

Layer TCP/IP	Keterangan
Application Layer	Lapisan yang berkaitan dengan layanan/protokol aplikasi.
Host-to-Host Layer	Lapisan yang menangani komunikasi antarsistem/host.
Internetwork Layer	Lapisan yang menangani komunikasi dan pengiriman paket antarjaringan.
Network Interface Layer	Lapisan yang berhubungan dengan interface dan media jaringan.

3. Wireshark

Wireshark adalah aplikasi penganalisa paket jaringan (Network Packet Analyzer) yang menangkap paket jaringan dan menampilkan data paket secara rinci. Wireshark bersifat open source dan dapat digunakan pada UNIX maupun Windows. Fitur yang dijelaskan dalam modul meliputi capture langsung dari network interface, tampilan informasi protokol secara rinci, penyimpanan/pembukaan data capture, serta filtering paket.

- Source: alamat IP pengirim.
- Destination: alamat IP tujuan.
- Protocol: jenis protokol yang digunakan.
- Length: panjang data yang dikirim.
- Info: informasi isi/aktivitas data yang dikirimkan.

4. Identifikasi Layer melalui Wireshark

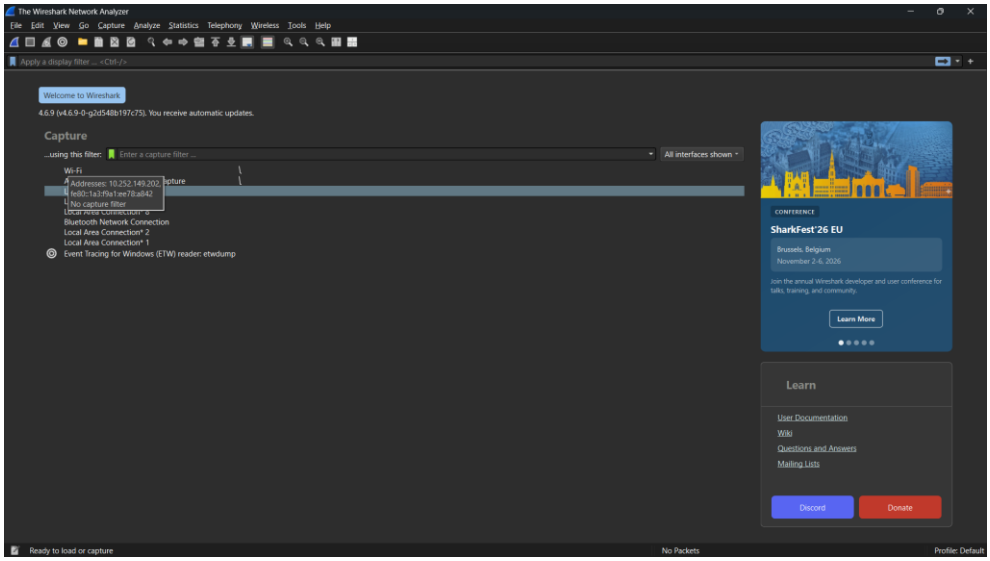
Pada contoh capture ICMP dalam modul, detail paket memperlihatkan Network Interface Layer/Ethernet II, Internetwork Layer/IPv4, dan Internet Control Message Protocol (ICMP). Hal ini menunjukkan bahwa Wireshark dapat digunakan untuk mengamati struktur data jaringan dan menghubungkannya dengan layer TCP/IP.

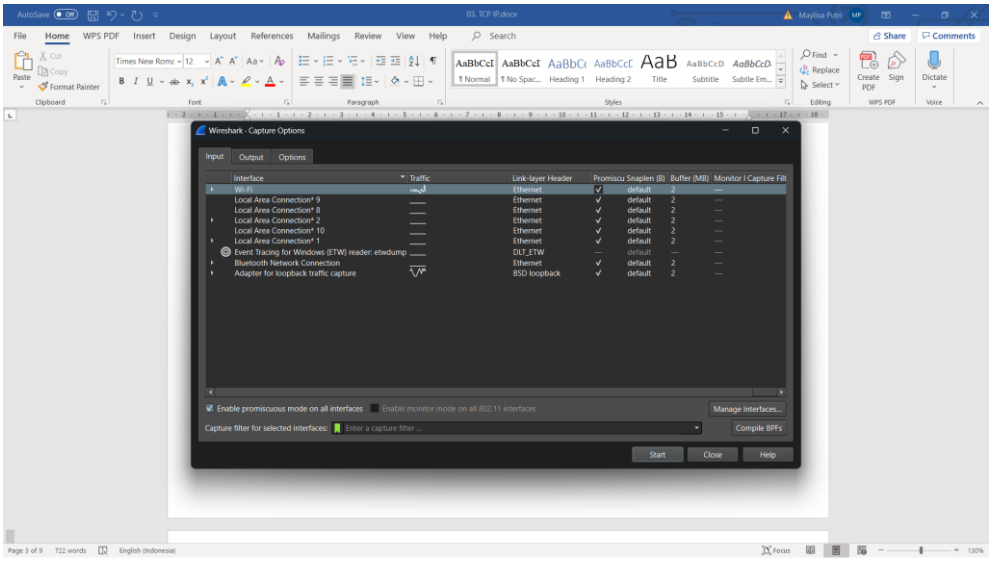
D. TUGAS

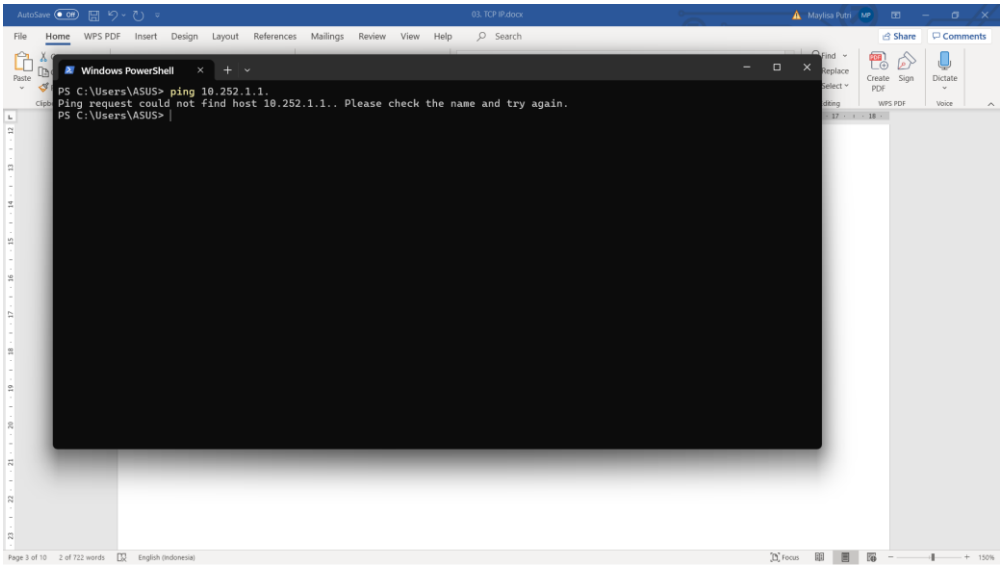
Lakukan capture menggunakan Wireshark, kemudian laporkan data yang direkam sesuai layer pada TCP/IP Model.

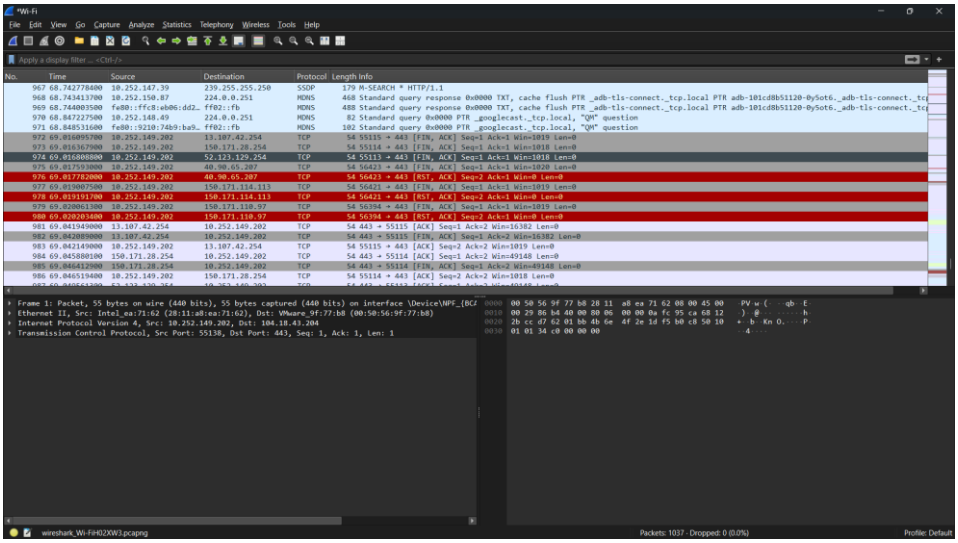
ICMP / PING

1. Buka Wireshark → Capture → Interfaces.
2. Pilih interface yang digunakan lalu klik Start.
3. Pada terminal jalankan ping 10.252.1.1.
4. Hentikan capture setelah proses selesai.
5. Gunakan filter icmp dan ambil screenshot hasilnya.


The Wireshark Network Analyzer
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
Welcome to Wireshark
4.6.9 (v4.6.9-0-g2d548b197c75). You receive automatic updates.
Capture
...using this filter: Enter a capture filter ...
All interfaces shown
Wi-Fi
Addresses: 10.252.149.202
fe80::1a3:f9a1:ee78:a842
No capture filter
Bluetooth Network Connection
Local Area Connection* 2
Local Area Connection* 1
Event Tracing for Windows (ETW) reader: etwdump
CONFERENCE
SharkFest'26 EU
Brussels, Belgium
November 2-6, 2026
Join the annual Wireshark developer and user conference for talks, training, and community.
Learn More
Learn
User Documentation
Wiki
Questions and Answers
Mailing Lists
Discord
Donate
Ready to load or capture
No Packets
Profile: Default


Wireshark - Capture Options
Input
Output
Options
Interface
Traffic
Link-layer Header
Promiscu
Snapshen (B)
Buffer (MB)
Monitor I
Capture Filt
Wi-Fi
Local Area Connection* 9
Local Area Connection* 8
Local Area Connection* 2
Local Area Connection* 10
Local Area Connection* 1
Event Tracing for Windows (ETW) reader: etwdump
Bluetooth Network Connection
Adapter for loopback traffic capture
Ethernet
DLT_ETW
BSD loopback
default
Enable promiscuous mode on all interfaces
Enable monitor mode on all 802.11 interfaces
Manage Interfaces...
Capture filter for selected interfaces:
Enter a capture filter ...
Compile BPFs
Start
Close
Help


Windows PowerShell
PS C:\Users\ASUS> ping 10.252.1.1.
Ping request could not find host 10.252.1.1.. Please check the name and try again.
PS C:\Users\ASUS>



Application Layer
Aktivitas yang dilakukan: PING menggunakan perintah ping 10.252.1.1.
Tidak ada protokol aplikasi HTTP/DNS/FTP pada paket ICMP yang dipilih.

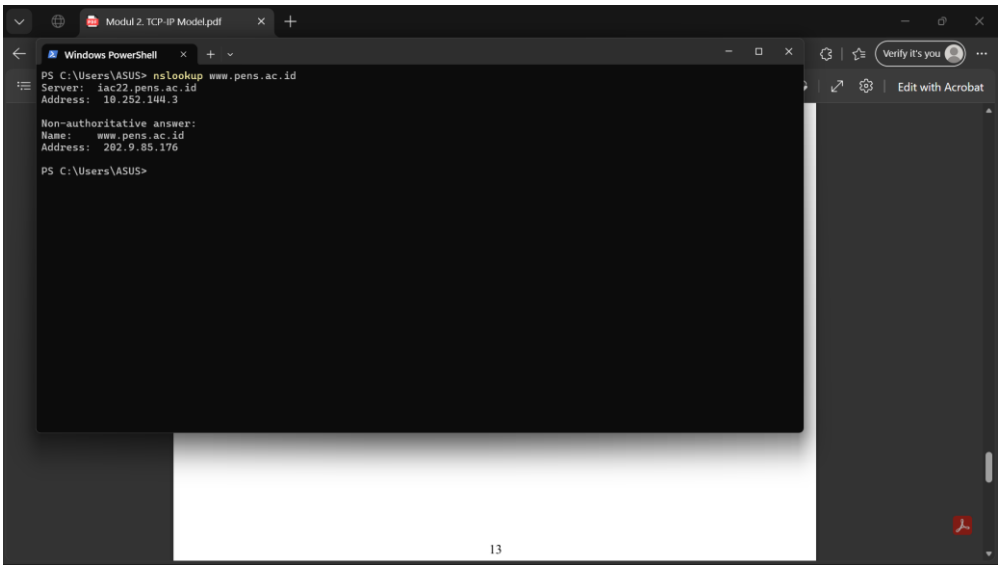
Host-to-Host Layer
ICMP tidak menggunakan TCP atau UDP; paket ICMP dibawa langsung oleh IPv4.
Source Port/Destination Port: tidak berlaku untuk paket ICMP.

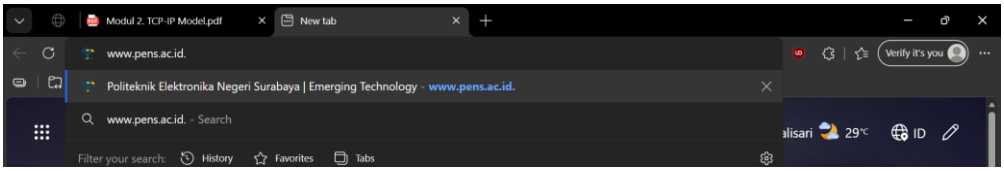
Internetwork Layer
Protocol: ICMP (Internet Control Message Protocol).
Source IP: 10.252.149.202
Destination IP: 52.123.129.254

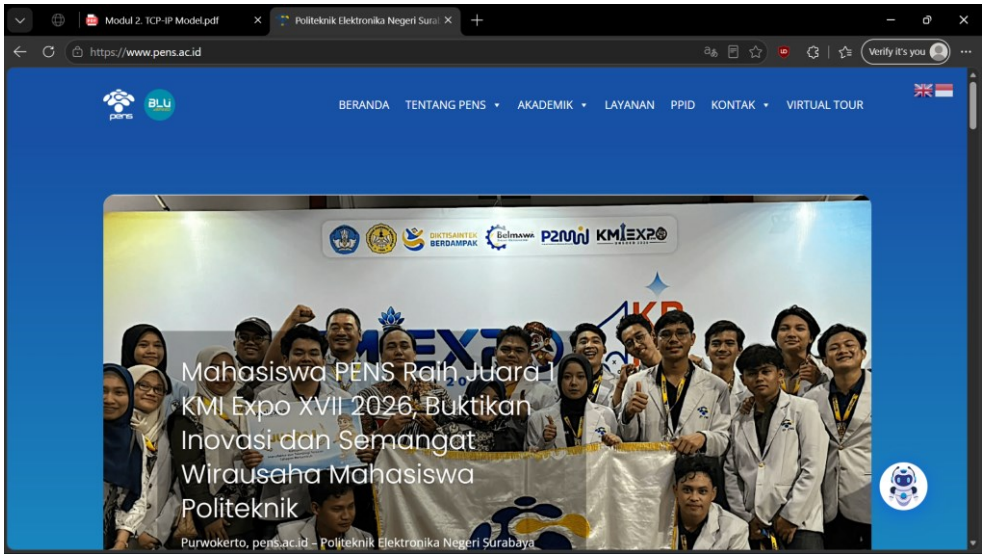
Network Interface Layer
Ethernet II.
Source MAC Address: 00:50:56:9f:77:b8.
Destination MAC Address: 28:11:a8:ea:71:62.

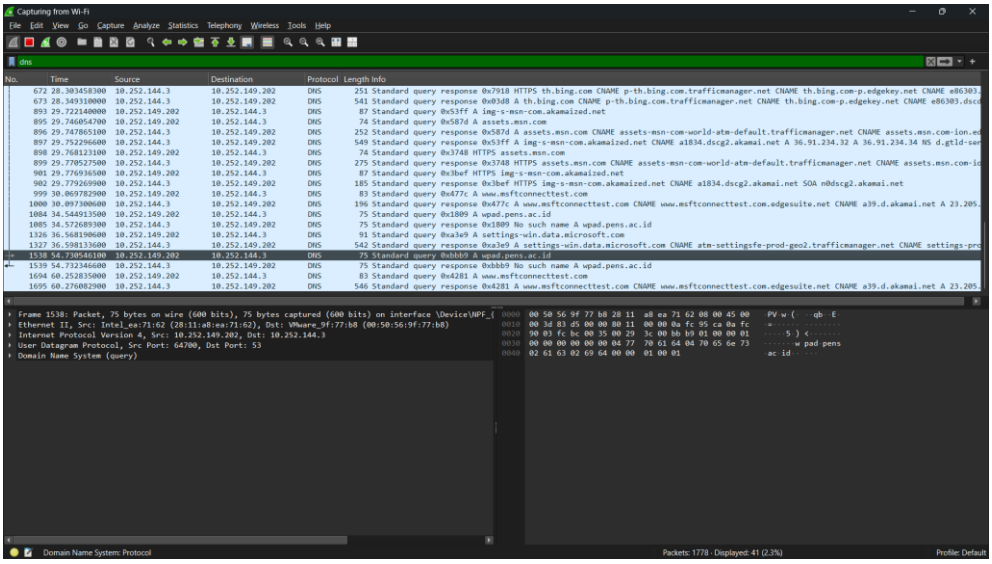
DNS / NSLOOKUP

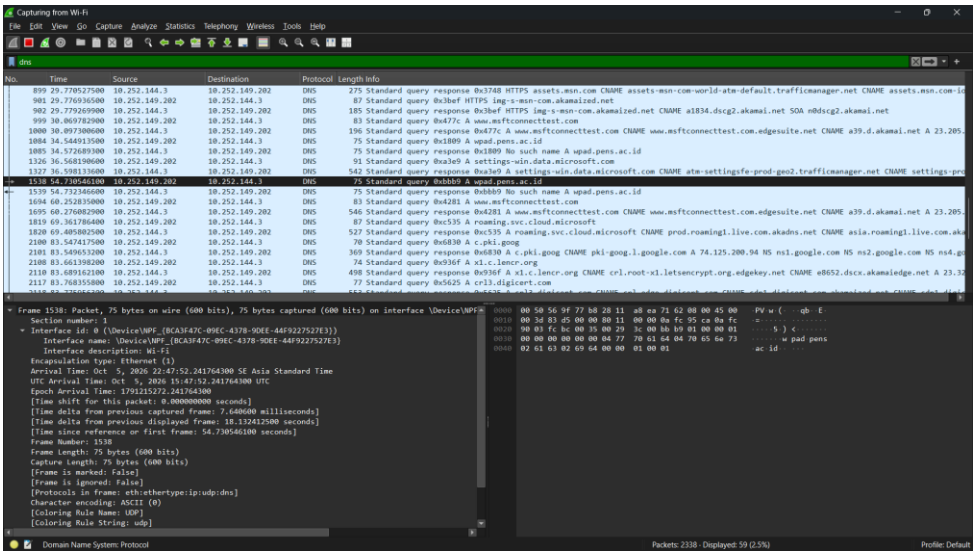
1. Mulai capture seperti kegiatan sebelumnya.
2. Jalankan nslookup www.pens.ac.id.
3. Hentikan capture dan gunakan filter dns.
4. Catat/screenshot data hasil capture.

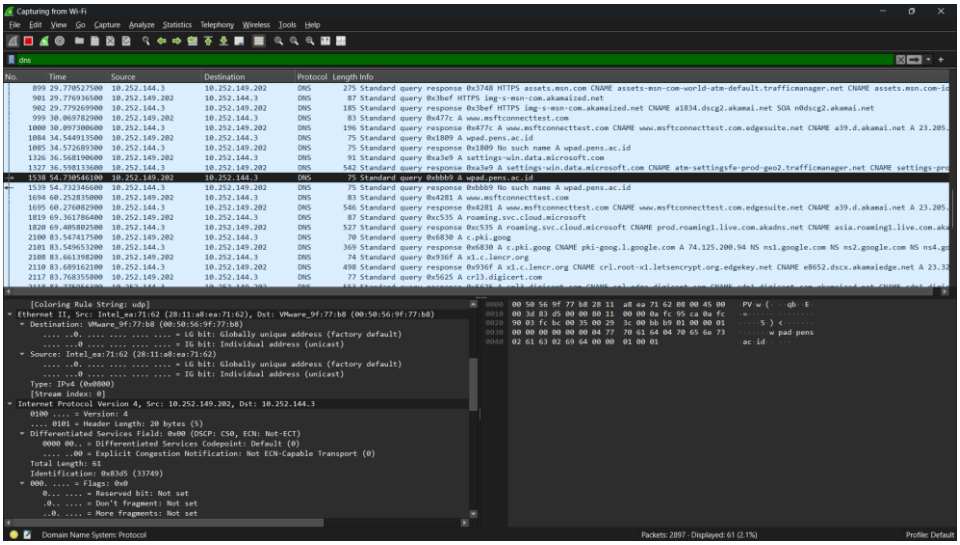


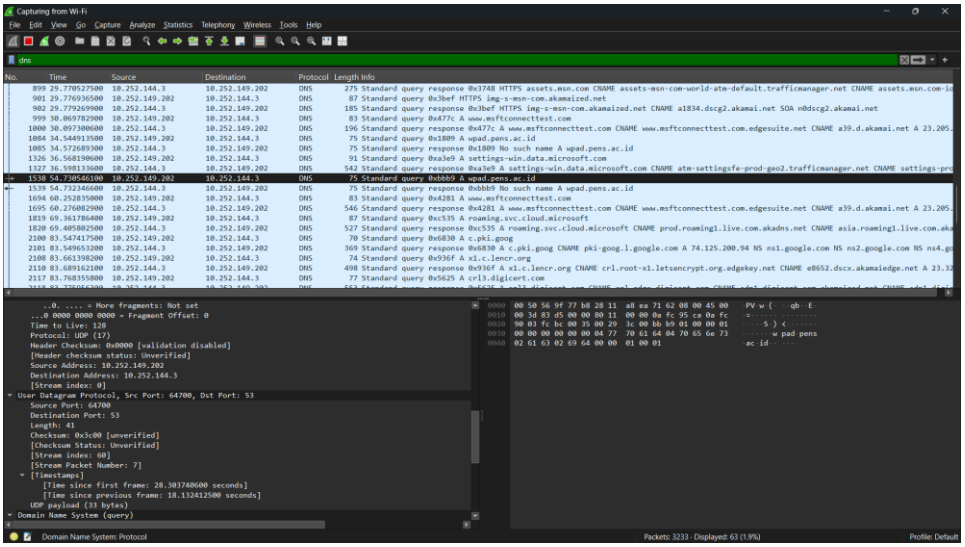


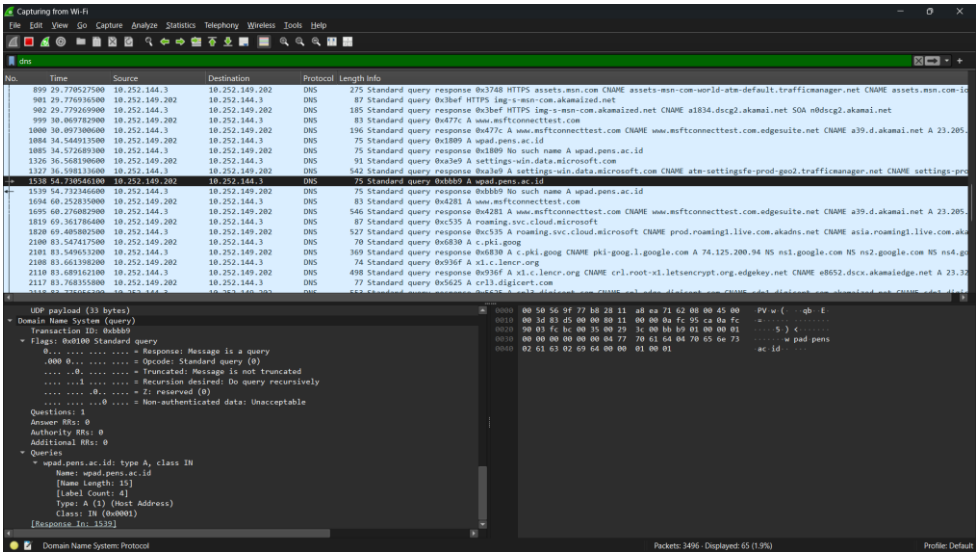



Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
dns
No. Time Source Destination Protocol Length Info
672 28.303458300 10.252.144.3 10.252.149.202 DNS 251 Standard query response 0x7918 HTTPS th.bing.com CNAME p-th.bing.com.trafficmanager.net CNAME th.bing.com-p.edgekey.net CNAME e86303.
673 28.349310000 10.252.144.3 10.252.149.202 DNS 541 Standard query response 0x03d8 A th.bing.com CNAME p-th.bing.com.trafficmanager.net CNAME th.bing.com-p.edgekey.net CNAME e86303.dscd
893 29.722140000 10.252.149.202 10.252.144.3 DNS 87 Standard query 0x53ff A img-s-msn-com.akamaized.net
895 29.746054700 10.252.149.202 10.252.144.3 DNS 74 Standard query 0x587d A assets.msn.com
896 29.747865100 10.252.144.3 10.252.149.202 DNS 252 Standard query response 0x587d A assets.msn.com CNAME assets-msn-com-world-atm-default.trafficmanager.net CNAME assets.msn.com-ion.ed
897 29.752296600 10.252.144.3 10.252.149.202 DNS 549 Standard query response 0x53ff A img-s-msn-com.akamaized.net CNAME a1834.dscg2.akamai.net A 36.91.234.32 A 36.91.234.34 NS d.gtld-ser
898 29.768123100 10.252.149.202 10.252.144.3 DNS 74 Standard query 0x3748 HTTPS assets.msn.com
899 29.770527500 10.252.144.3 10.252.149.202 DNS 275 Standard query response 0x3748 HTTPS assets.msn.com CNAME assets-msn-com-world-atm-default.trafficmanager.net CNAME assets.msn.com-ic
901 29.776936500 10.252.149.202 10.252.144.3 DNS 87 Standard query 0x3bef HTTPS img-s-msn-com.akamaized.net
902 29.779269900 10.252.144.3 10.252.149.202 DNS 185 Standard query response 0x3bef HTTPS img-s-msn-com.akamaized.net CNAME a1834.dscg2.akamai.net SOA n0dscg2.akamai.net
999 30.069782900 10.252.149.202 10.252.144.3 DNS 83 Standard query 0x477c A www.msftconnecttest.com
1000 30.097300600 10.252.144.3 10.252.149.202 DNS 196 Standard query response 0x477c A www.msftconnecttest.com CNAME www.msftconnecttest.com.edgesuite.net CNAME a39.d.akamai.net A 23.205.
1084 34.544913500 10.252.149.202 10.252.144.3 DNS 75 Standard query 0x1809 A wpad.pens.ac.id
1085 34.572689300 10.252.144.3 10.252.149.202 DNS 75 Standard query response 0x1809 No such name A wpad.pens.ac.id
1326 36.568190600 10.252.149.202 10.252.144.3 DNS 91 Standard query 0xa3e9 A settings-win.data.microsoft.com
1327 36.598133600 10.252.144.3 10.252.149.202 DNS 542 Standard query response 0xa3e9 A settings-win.data.microsoft.com CNAME atm-settingsfe-prod-geo2.trafficmanager.net CNAME settings-pro
1538 54.730546100 10.252.149.202 10.252.144.3 DNS 75 Standard query 0xbbb9 A wpad.pens.ac.id
1539 54.732346600 10.252.144.3 10.252.149.202 DNS 75 Standard query response 0xbbb9 No such name A wpad.pens.ac.id
1694 60.252835000 10.252.149.202 10.252.144.3 DNS 83 Standard query 0x4281 A www.msftconnecttest.com
1695 60.276082900 10.252.144.3 10.252.149.202 DNS 546 Standard query response 0x4281 A www.msftconnecttest.com CNAME www.msftconnecttest.com.edgesuite.net CNAME a39.d.akamai.net A 23.205.
Frame 1538: Packet, 75 bytes on wire (600 bits), 75 bytes captured (600 bits) on interface \Device\NPF_{
Ethernet II, Src: Intel_ea:71:62 (28:11:a8:ea:71:62), Dst: VMware_9f:77:b8 (00:50:56:9f:77:b8)
Internet Protocol Version 4, Src: 10.252.149.202, Dst: 10.252.144.3
User Datagram Protocol, Src Port: 64700, Dst Port: 53
Domain Name System (query)
0000 00 50 56 9f 77 b8 28 11 a8 ea 71 62 08 00 45 00 PV w (qb E
0010 00 3d 83 d5 00 00 80 11 00 00 0a fc 95 ca 0a fc =
0020 90 03 fc bc 00 35 00 29 3c 00 bb b9 01 00 00 01 5) <
0030 00 00 00 00 00 00 04 77 70 61 64 04 70 65 6e 73 w pad pens
0040 02 61 63 02 69 64 00 00 01 00 01 ac id
Domain Name System: Protocol
Packets: 1778 · Displayed: 41 (2.3%)
Profile: Default


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
dns
No. Time Source Destination Protocol Length Info
899 29.770527500 10.252.144.3 10.252.149.202 DNS 275 Standard query response 0x3748 HTTPS assets.msn.com CNAME assets-msn-com-world-atm-default.trafficmanager.net CNAME assets.msn.com-ic
901 29.776936500 10.252.149.202 10.252.144.3 DNS 87 Standard query 0x3bef HTTPS img-s-msn-com.akamaized.net
902 29.779269900 10.252.144.3 10.252.149.202 DNS 185 Standard query response 0x3bef HTTPS img-s-msn-com.akamaized.net CNAME a1834.dscg2.akamai.net SOA n0dscg2.akamai.net
999 30.069782900 10.252.149.202 10.252.144.3 DNS 83 Standard query 0x477c A www.msftconnecttest.com
1000 30.097300600 10.252.144.3 10.252.149.202 DNS 196 Standard query response 0x477c A www.msftconnecttest.com CNAME www.msftconnecttest.com.edgesuite.net CNAME a39.d.akamai.net A 23.205.
1084 34.544913500 10.252.149.202 10.252.144.3 DNS 75 Standard query 0x1809 A wpad.pens.ac.id
1085 34.572689300 10.252.144.3 10.252.149.202 DNS 75 Standard query response 0x1809 No such name A wpad.pens.ac.id
1326 36.568190600 10.252.149.202 10.252.144.3 DNS 91 Standard query 0xa3e9 A settings-win.data.microsoft.com
1327 36.598133600 10.252.144.3 10.252.149.202 DNS 542 Standard query response 0xa3e9 A settings-win.data.microsoft.com CNAME atm-settingsfe-prod-geo2.trafficmanager.net CNAME settings-pro
1538 54.730546100 10.252.149.202 10.252.144.3 DNS 75 Standard query 0xbbb9 A wpad.pens.ac.id
1539 54.732346600 10.252.144.3 10.252.149.202 DNS 75 Standard query response 0xbbb9 No such name A wpad.pens.ac.id
1694 60.252835000 10.252.149.202 10.252.144.3 DNS 83 Standard query 0x4281 A www.msftconnecttest.com
1695 60.276082900 10.252.144.3 10.252.149.202 DNS 546 Standard query response 0x4281 A www.msftconnecttest.com CNAME www.msftconnecttest.com.edgesuite.net CNAME a39.d.akamai.net A 23.205.
1819 69.361786400 10.252.149.202 10.252.144.3 DNS 87 Standard query 0xc535 A roaming.svc.cloud.microsoft
1820 69.405802500 10.252.144.3 10.252.149.202 DNS 527 Standard query response 0xc535 A roaming.svc.cloud.microsoft CNAME prod.roaming1.live.com.akadns.net CNAME asia.roaming1.live.com.aka
2100 83.547417500 10.252.149.202 10.252.144.3 DNS 70 Standard query 0x6830 A c.pki.goog
2101 83.549653200 10.252.144.3 10.252.149.202 DNS 369 Standard query response 0x6830 A c.pki.goog CNAME pki-goog.l.google.com A 74.125.200.94 NS ns1.google.com NS ns2.google.com NS ns4.g
2108 83.661398200 10.252.149.202 10.252.144.3 DNS 74 Standard query 0x936f A x1.c.lencr.org
2110 83.689162100 10.252.144.3 10.252.149.202 DNS 498 Standard query response 0x936f A x1.c.lencr.org CNAME crl.root-x1.letsencrypt.org.edgekey.net CNAME e8652.dscx.akamaiedge.net A 23.32
2117 83.768355800 10.252.149.202 10.252.144.3 DNS 77 Standard query 0x5625 A crl3.digicert.com
Frame 1538: Packet, 75 bytes on wire (600 bits), 75 bytes captured (600 bits) on interface \Device\NPF_{
Section number: 1
Interface id: 0 (\Device\NPF_{BCA3F47C-09EC-4378-9DEE-44F9227527E3})
Interface name: \Device\NPF_{BCA3F47C-09EC-4378-9DEE-44F9227527E3}
Interface description: Wi-Fi
Encapsulation type: Ethernet (1)
Arrival Time: Oct 5, 2026 22:47:52.241764300 SE Asia Standard Time
UTC Arrival Time: Oct 5, 2026 15:47:52.241764300 UTC
Epoch Arrival Time: 1791215272.241764300
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 7.640600 milliseconds]
[Time delta from previous displayed frame: 18.132412500 seconds]
[Time since reference or first frame: 54.730546100 seconds]
Frame Number: 1538
Frame Length: 75 bytes (600 bits)
Capture Length: 75 bytes (600 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:udp:dns]
Character encoding: ASCII (0)
[Coloring Rule Name: UDP]
[Coloring Rule String: udp]
0000 00 50 56 9f 77 b8 28 11 a8 ea 71 62 08 00 45 00 PV w (qb E
0010 00 3d 83 d5 00 00 80 11 00 00 0a fc 95 ca 0a fc =
0020 90 03 fc bc 00 35 00 29 3c 00 bb b9 01 00 00 01 5) <
0030 00 00 00 00 00 00 04 77 70 61 64 04 70 65 6e 73 w pad pens
0040 02 61 63 02 69 64 00 00 01 00 01 ac id
Domain Name System: Protocol
Packets: 2338 · Displayed: 59 (2.5%)
Profile: Default


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
dns
No. Time Source Destination Protocol Length Info
899 29.770527500 10.252.144.3 10.252.149.202 DNS 275 Standard query response 0x3748 HTTPS assets.msn.com CNAME assets-msn-com-world-atm-default.trafficmanager.net CNAME assets.msn.com-ic
901 29.776936500 10.252.149.202 10.252.144.3 DNS 87 Standard query 0x3bef HTTPS img-s-msn-com.akamaized.net
902 29.779269900 10.252.144.3 10.252.149.202 DNS 185 Standard query response 0x3bef HTTPS img-s-msn-com.akamaized.net CNAME a1834.dscg2.akamai.net SOA n0dscg2.akamai.net
999 30.069782900 10.252.149.202 10.252.144.3 DNS 83 Standard query 0x477c A www.msftconnecttest.com
1000 30.097300600 10.252.144.3 10.252.149.202 DNS 196 Standard query response 0x477c A www.msftconnecttest.com CNAME www.msftconnecttest.com.edgesuite.net CNAME a39.d.akamai.net A 23.205.
1084 34.544913500 10.252.149.202 10.252.144.3 DNS 75 Standard query 0x1809 A wpad.pens.ac.id
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1326 36.568190600 10.252.149.202 10.252.144.3 DNS 91 Standard query 0xa3e9 A settings-win.data.microsoft.com
1327 36.598133600 10.252.144.3 10.252.149.202 DNS 542 Standard query response 0xa3e9 A settings-win.data.microsoft.com CNAME atm-settingsfe-prod-geo2.trafficmanager.net CNAME settings-pro
1538 54.730546100 10.252.149.202 10.252.144.3 DNS 75 Standard query 0xbbb9 A wpad.pens.ac.id
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1819 69.361786400 10.252.149.202 10.252.144.3 DNS 87 Standard query 0xc535 A roaming.svc.cloud.microsoft
1820 69.405802500 10.252.144.3 10.252.149.202 DNS 527 Standard query response 0xc535 A roaming.svc.cloud.microsoft CNAME prod.roaming1.live.com.akadns.net CNAME asia.roaming1.live.com.aka
2100 83.547417500 10.252.149.202 10.252.144.3 DNS 70 Standard query 0x6830 A c.pki.goog
2101 83.549653200 10.252.144.3 10.252.149.202 DNS 369 Standard query response 0x6830 A c.pki.goog CNAME pki-goog.l.google.com A 74.125.200.94 NS ns1.google.com NS ns2.google.com NS ns4.g
2108 83.661398200 10.252.149.202 10.252.144.3 DNS 74 Standard query 0x936f A x1.c.lencr.org
2110 83.689162100 10.252.144.3 10.252.149.202 DNS 498 Standard query response 0x936f A x1.c.lencr.org CNAME crl.root-x1.letsencrypt.org.edgekey.net CNAME e8652.dscx.akamaiedge.net A 23.32
2117 83.768355800 10.252.149.202 10.252.144.3 DNS 77 Standard query 0x5625 A crl3.digicert.com
[Coloring Rule String: udp]
Ethernet II, Src: Intel_ea:71:62 (28:11:a8:ea:71:62), Dst: VMware_9f:77:b8 (00:50:56:9f:77:b8)
Destination: VMware_9f:77:b8 (00:50:56:9f:77:b8)
.... ..0. .... = LG bit: Globally unique address (factory default)
....0 .... = IG bit: Individual address (unicast)
Source: Intel_ea:71:62 (28:11:a8:ea:71:62)
.... ..0. .... = LG bit: Globally unique address (factory default)
....0 .... = IG bit: Individual address (unicast)
Type: IPv4 (0x0800)
[Stream index: 0]
Internet Protocol Version 4, Src: 10.252.149.202, Dst: 10.252.144.3
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
0000 00.. = Differentiated Services Codepoint: Default (0)
.... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
Total Length: 61
Identification: 0x83d5 (33749)
000. .... = Flags: 0x0
0... .... = Reserved bit: Not set
.0.. .... = Don't fragment: Not set
..0. .... = More fragments: Not set
0000 00 50 56 9f 77 b8 28 11 a8 ea 71 62 08 00 45 00 PV w (qb E
0010 00 3d 83 d5 00 00 80 11 00 00 0a fc 95 ca 0a fc =
0020 90 03 fc bc 00 35 00 29 3c 00 bb b9 01 00 00 01 5) <
0030 00 00 00 00 00 00 04 77 70 61 64 04 70 65 6e 73 w pad pens
0040 02 61 63 02 69 64 00 00 01 00 01 ac id
Domain Name System: Protocol
Packets: 2897 · Displayed: 61 (2.1%)
Profile: Default





Application Layer
DNS (Domain Name System).
Aktivitas: nslookup www.pens.ac.id.
Query name: www.pens.ac.id.
Jenis pesan: DNS query atau DNS response; tulis sesuai paket yang dipilih.

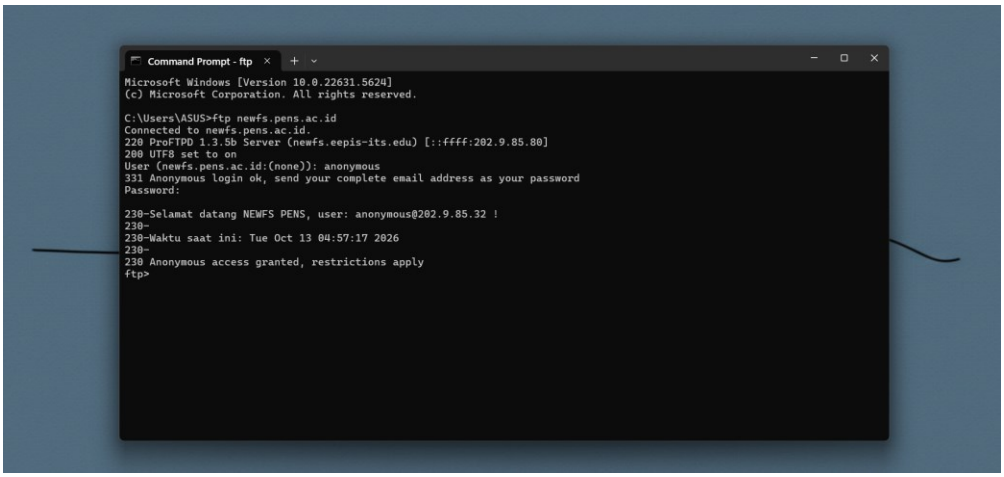
Host-to-Host Layer
UDP
Port DNS: 53; catat Source Port dan Destination Port persis seperti detail paket Wireshark.

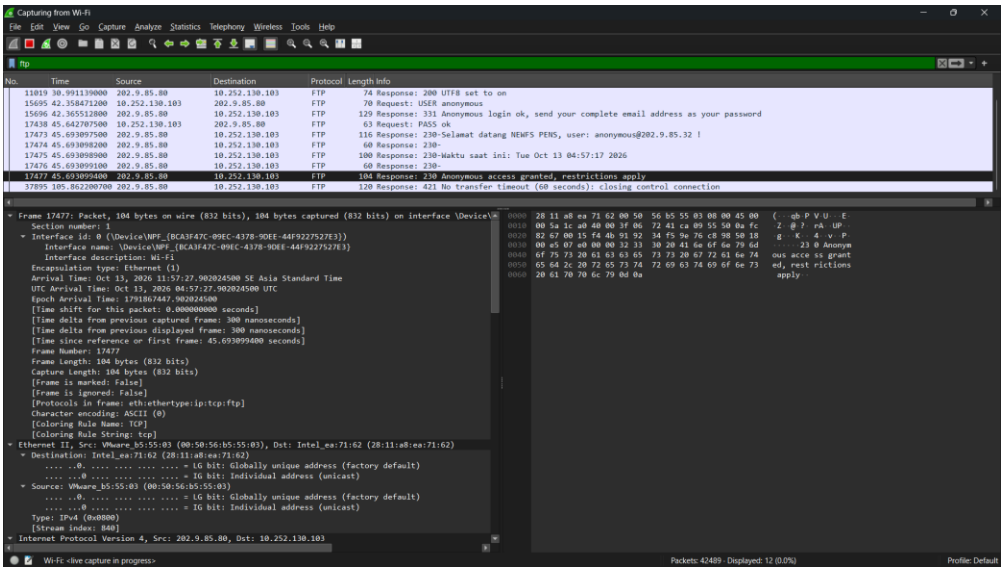
Internetwork Layer
IPv4 atau IPv6, sesuai paket yang dipilih.
Source IP: 10.252.149.202
Destination IP: 10.252.144.3

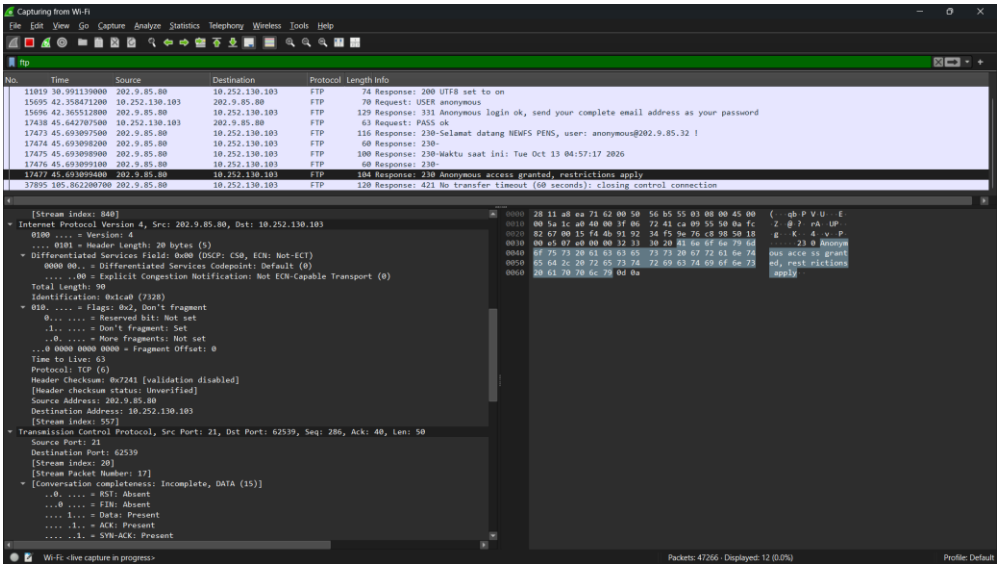
Network Interface Layer
Ethernet II.
Source MAC Address: 00:50:56:9f:77:b8.
Destination MAC Address: 28:11:a8:ea:71:62.

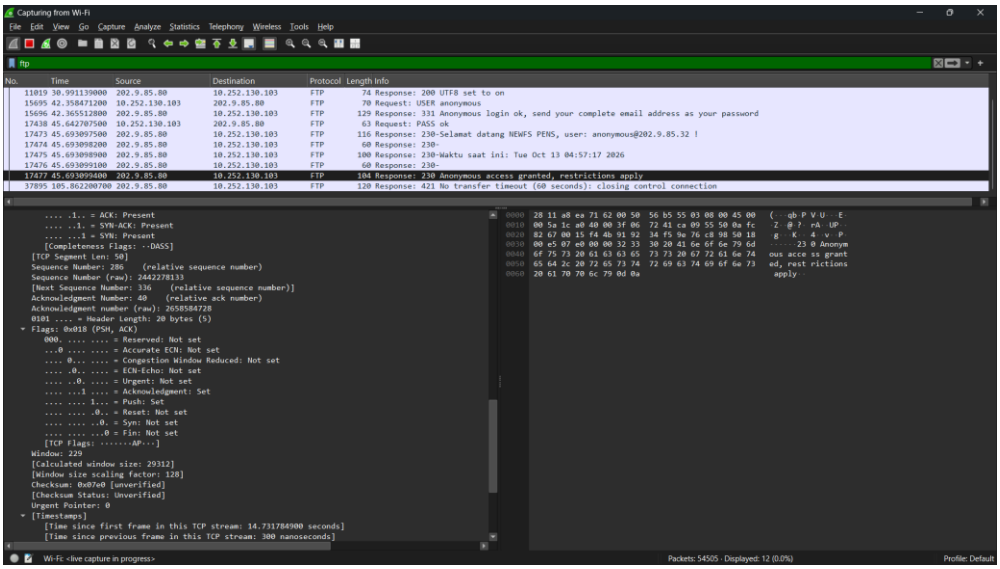
FTP

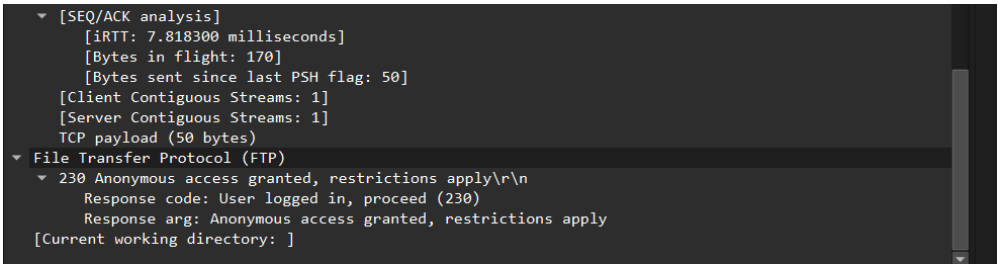
1. Mulai capture.
2. Buka ftp://newfs.pens.ac.id melalui browser.
3. Download salah satu data yang tersedia.
4. Hentikan capture, gunakan filter ftp, dan dokumentasikan hasilnya.











Application Layer
FTP (File Transfer Protocol).
Aktivitas: membuka ftp://newfs.pens.ac.id dan mencoba mengunduh salah satu file.
Screenshot menunjukkan respons FTP seperti kode 220 (service ready) dan 230 (anonymous access granted), sesuai paket yang ditampilkan.

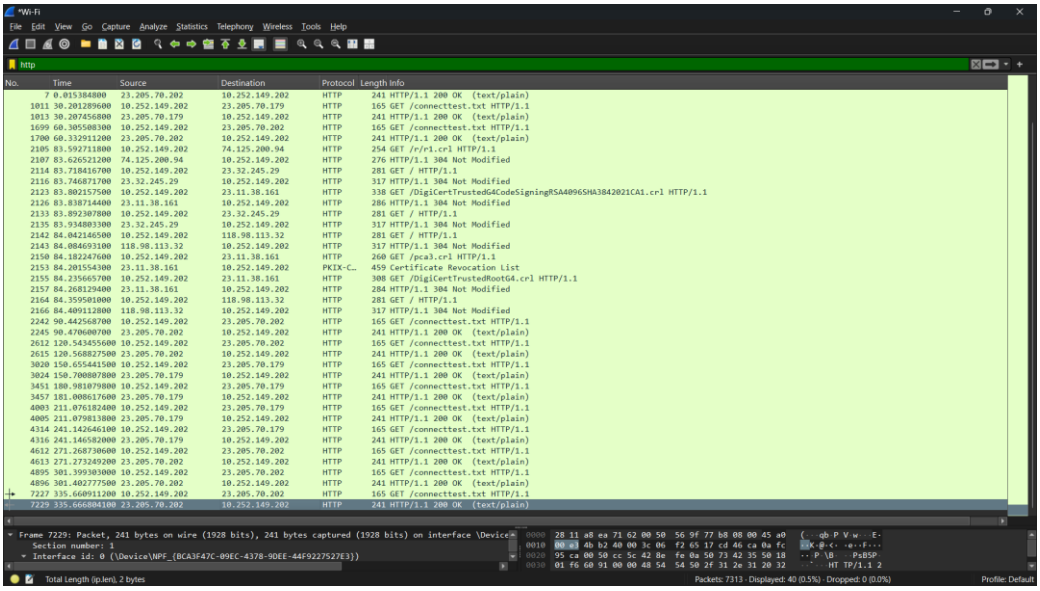
Host-to-Host Layer
TCP (Transmission Control Protocol).
FTP control connection biasanya menggunakan port 21 pada server; tulis Source Port, Destination Port, dan flags sesuai paket yang dipilih.

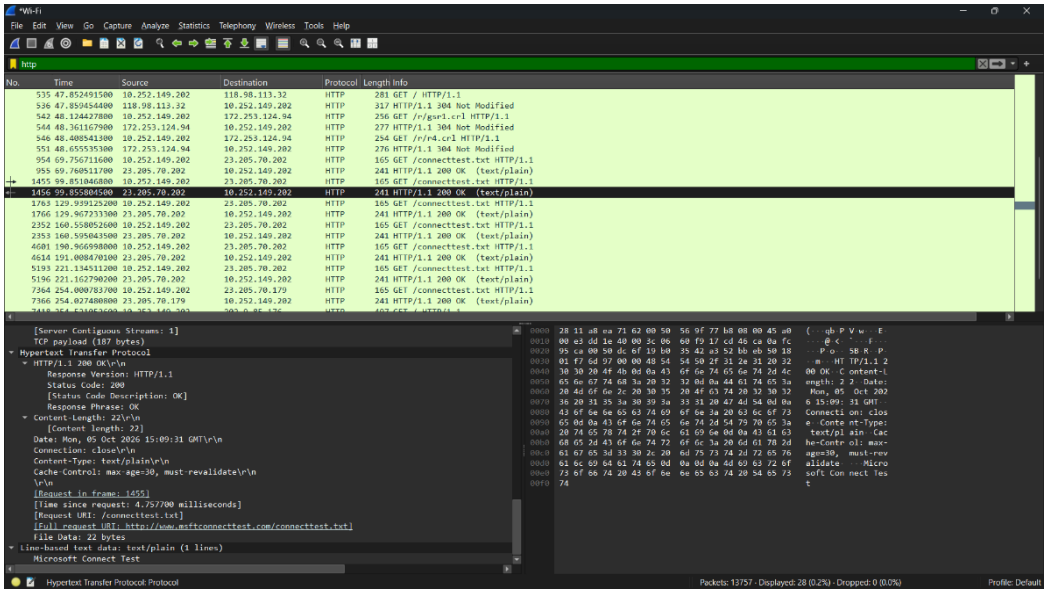
Internetwork Layer
IPv4 atau IPv6, sesuai detail paket.
Source IP: 202.9.85.80
Destination IP: 10.252.130.103

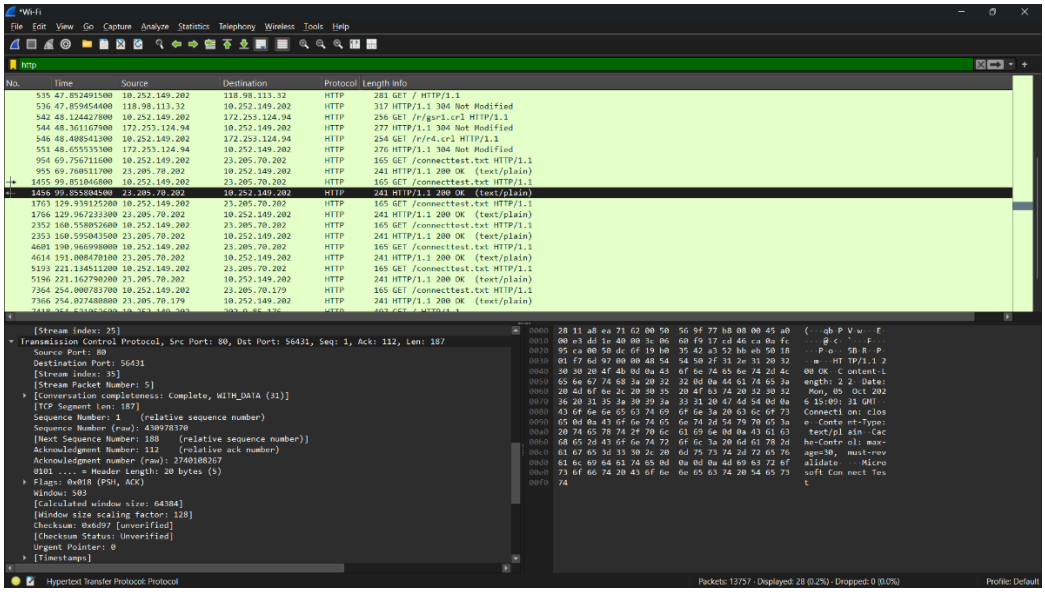
Network Interface Layer
Ethernet II.
Source MAC Address: 00:50:56:9f:77:b8.
Destination MAC Address: 28:11:a8:ea:71:62.

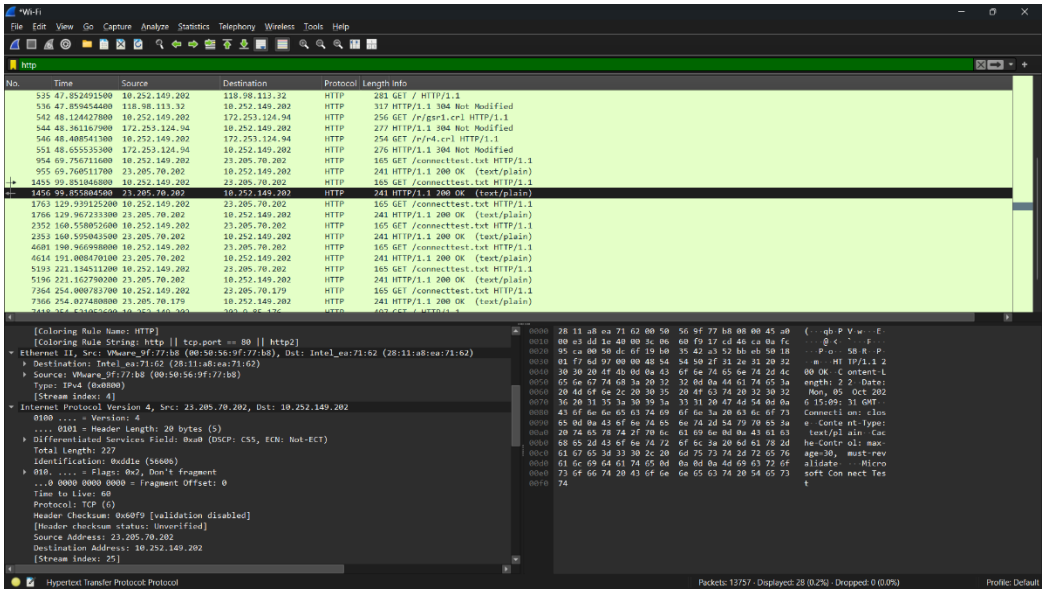
HTTP

1. Mulai capture.
2. Buka http://www.pens.ac.id melalui browser.
3. Hentikan capture setelah aktivitas terekam.
4. Gunakan filter http dan dokumentasikan hasilnya.









Application Layer
HTTP (Hypertext Transfer Protocol).
Response: HTTP/1.1 200 OK.
Request URI: /connecttest.txt.
Full Request URI: http://www.msftconnecttest.com/connecttest.txt.
Content-Type: text/plain.
File Data: Microsoft Connect Test.

Host-to-Host Layer
TCP (Transmission Control Protocol).
Source Port: 80.
Destination Port: 56431.
TCP Segment Length: 187 bytes.
Flags: PSH, ACK.

Internetwork Layer
IPv4 (Internet Protocol Version 4).
Source IP: 23.205.70.202.
Destination IP: 10.252.149.202.
Header Length: 20 bytes.
Total Length: 227 bytes.
Time to Live (TTL): 60.
Protocol: TCP (6).

Network Interface Layer
Ethernet II.
Source MAC Address: 00:50:56:9f:77:b8.
Destination MAC Address: 28:11:a8:ea:71:62.

E. KESIMPULAN

Berdasarkan praktikum TCP/IP Model menggunakan aplikasi Wireshark, dapat disimpulkan bahwa proses komunikasi data dalam jaringan dapat diamati dan dianalisis melalui empat layer TCP/IP, yaitu Application Layer, Host-to-Host Layer, Internetwork Layer, dan Network Interface Layer. Melalui kegiatan ICMP/PING, DNS/NSLOOKUP, FTP, dan HTTP, setiap protokol dapat diidentifikasi berdasarkan fungsi dan informasi paket yang ditampilkan pada Wireshark. ICMP digunakan untuk menguji konektivitas jaringan, DNS untuk mengetahui alamat IP berdasarkan nama domain, FTP untuk melakukan transfer file, dan HTTP untuk komunikasi dalam mengakses data web. Dengan demikian, praktikum ini membantu mahasiswa memahami proses transmisi data serta mengidentifikasi protokol dan layer yang terlibat dalam komunikasi jaringan.

F. REFERENSI

Dhoto. Modul Praktikum Jaringan Komputer – BAB 2: TCP/IP Model. Politeknik Elektronika Negeri Surabaya (PENS).